WORKED SOLUTION

STEP II 1991, Question 6

Pure • Bounding Integrals

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q6

The key idea

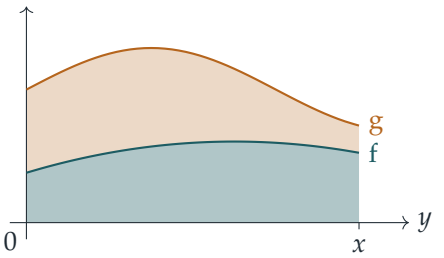
An integral from 0 to x is an area, and a smaller non-negative function encloses a smaller area. That one fact, used over and over, turns a known inequality into a new one. Integrating $0 \leq \cos y \leq 1$ gives a bound on $\sin x$; integrating that gives a bound on $\cos x$; integrating again gives a sharper bound on $\sin x$. Each integral bound then follows by trapping the integrand between two functions we can integrate exactly.

Solution

Comparing integrals

Since $\mathrm{f}(y) \geq 0$ on $[0, x]$, the graph of f lies on or above the y-axis, so $\int_0^x \mathrm{f}(y)\,\mathrm{d}y$ is the area between the graph and the axis, which is non-negative. Since $\mathrm{f}(y) \leq \mathrm{g}(y)$, the region under f sits inside the region under g, so its area is no larger:

$$0 \leq \int_0^x \mathrm{f}(y)\,\mathrm{d}y \leq \int_0^x \mathrm{g}(y)\,\mathrm{d}y.$$



(Analytically: $\mathrm{g} - \mathrm{f} \geq 0$, so $\int_0^x (\mathrm{g} - \mathrm{f})\,\mathrm{d}y \geq 0$, and the integral is linear.)

The bounds on $\sin x$ and $\cos x$

Take $0 \le x \le \frac{1}{2}\pi$. For $0 \le y \le x$ we have $0 \le \cos y \le 1$, so by the result above with $\mathrm{f} = \cos$, $\mathrm{g} = 1$,

$$0 \le \int_0^x \cos y \, \mathrm{d}y \le \int_0^x 1 \, \mathrm{d}y, \qquad \text{that is} \qquad 0 \le \sin x \le x.$$

This holds for every x in $[0, \frac{1}{2}\pi]$, so in particular $0 \le \sin y \le y$ for $0 \le y \le x$. Applying the result again with $\mathrm{f} = \sin$, $\mathrm{g}(y) = y$:

$$\int_0^x \sin y \, \mathrm{d}y \le \int_0^x y \, \mathrm{d}y \quad \Longrightarrow \quad 1 - \cos x \le \tfrac{1}{2}x^2 \quad \Longrightarrow \quad \cos x \ge 1 - \tfrac{1}{2}x^2.$$

The first integral

On $[0, \frac{1}{10}]$ (well inside $[0, \frac{1}{2}\pi]$) we now have $1 - \frac{1}{2}x^2 \le \cos x \le 1$, so

$$3 - \tfrac{1}{2}x^2 \le 2 + \cos x \le 3.$$

? **Ask yourself** Can we just square and take reciprocals?

Yes, because every quantity here is positive: $3 - \frac{1}{2}x^2 \ge 3 - \frac{1}{200} > 0$. For positive numbers, squaring keeps the order and taking reciprocals reverses it. Multiplying by $x \ge 0$ then gives

$$\frac{x}{9} \le \frac{x}{(2 + \cos x)^2} \le \frac{x}{(3 - \frac{1}{2}x^2)^2} \qquad (0 \le x \le \tfrac{1}{10}).$$

All three functions are non-negative, so we may integrate from 0 to $\frac{1}{10}$. The left gives

$$\int_0^{1/10} \frac{x}{9} \, \mathrm{d}x = \frac{1}{18} \cdot \frac{1}{100} = \frac{1}{1800}.$$

For the right, the numerator x is, up to sign, the derivative of $3 - \frac{1}{2}x^2$, and $\frac{\mathrm{d}}{\mathrm{d}x}(3 - \frac{1}{2}x^2)^{-1} = x(3 - \frac{1}{2}x^2)^{-2}$. So

$$\int_0^{1/10} \frac{x}{(3 - \frac{1}{2}x^2)^2} \, \mathrm{d}x = \left[\frac{1}{3 - \frac{1}{2}x^2}\right]_0^{1/10} = \frac{200}{599} - \frac{1}{3} = \frac{600 - 599}{1797} = \frac{1}{1797}.$$

Hence

$$\frac{1}{1800} \le \int_0^{1/10} \frac{x}{(2 + \cos x)^2} \, \mathrm{d}x \le \frac{1}{1797}.$$

A sharper bound on $\sin x$

For $0 \le y \le x \le \frac{1}{2}\pi$ we have $\cos y \ge 1 - \frac{1}{2}y^2$, i.e. $0 \le 1 - \cos y \le \frac{1}{2}y^2$ (the left inequality because $\cos y \le 1$). Integrating from 0 to x:

$$\int_0^x (1 - \cos y) \, \mathrm{d}y \le \int_0^x \tfrac{1}{2}y^2 \, \mathrm{d}y \quad \Longrightarrow \quad x - \sin x \le \tfrac{1}{6}x^3 \quad \Longrightarrow \quad \sin x \ge x - \tfrac{1}{6}x^3.$$

The second integral

On $[0, \frac{1}{10}]$ we have $x - \frac{1}{6}x^3 \le \sin x \le x$, so

$$1 - \tfrac{1}{6}x^3 \le 1 - x + \sin x \le 1.$$

Ask yourself Why is this the right combination to look at?

The $-x$ in the denominator is there to cancel the leading term of $\sin x$, leaving 1 plus something of order x^3. The lower bound $1 - \frac{1}{6}x^3 \ge 1 - \frac{1}{6000}$ is positive, so as before we square, take reciprocals (reversing the order) and multiply by $x^2 \ge 0$:

$$x^2 \le \frac{x^2}{(1 - x + \sin x)^2} \le \frac{x^2}{(1 - \frac{1}{6}x^3)^2}.$$

Integrating, the left gives $\int_0^{1/10} x^2\,\mathrm{d}x = \frac{1}{3} \cdot \frac{1}{1000} = \frac{1}{3000}$. For the right, $\frac{\mathrm{d}}{\mathrm{d}x}(1 - \frac{1}{6}x^3)^{-1} = \frac{1}{2}x^2(1 - \frac{1}{6}x^3)^{-2}$, so

$$\int_0^{1/10} \frac{x^2}{(1 - \frac{1}{6}x^3)^2}\,\mathrm{d}x = 2\left[\frac{1}{1 - \frac{1}{6}x^3}\right]_0^{1/10} = 2\left(\frac{6000}{5999} - 1\right) = \frac{2}{5999}.$$

Hence

$$\frac{1}{3000} \le \int_0^{1/10} \frac{x^2}{(1 - x + \sin x)^2}\,\mathrm{d}x \le \frac{2}{5999}.$$

Checks

- Simpson's rule gives $\int_0^{1/10} x(2 + \cos x)^{-2}\,\mathrm{d}x \approx 0.00055648$, between $1/1800 \approx 0.00055556$ and $1/1797 \approx 0.00055648$ (the upper bound is extremely tight: it agrees to about 5×10^{-10}).
- Similarly $\int_0^{1/10} x^2(1 - x + \sin x)^{-2}\,\mathrm{d}x \approx 0.000333389$, between $1/3000 \approx 0.000333333$ and $2/5999 \approx 0.000333389$.
- The three pointwise inequalities held at 10^5 random points of $[0, \frac{1}{2}\pi]$, with equality at $x = 0$.

Common slips

- Taking reciprocals without checking signs: $a \le b$ gives $1/a \ge 1/b$ only when a and b have the same sign.
- Forgetting that the comparison result needs f ≥ 0 for the left-hand inequality; here $\sin y \ge 0$ and $1 - \cos y \ge 0$ supply it.
- Using the bound $\cos x \le 1$ for the upper bound of the integral: $\cos x \le 1$ makes the denominator *smaller than* 9, which gives the lower bound $1/1800$, not the upper one.

Beyond STEP

Repeatedly integrating $0 \leq \cos y \leq 1$ produces the whole chain of alternating Taylor bounds $x - \frac{1}{6}x^3 \leq \sin x \leq x - \frac{1}{6}x^3 + \frac{1}{120}x^5 \leq \cdots$, which is one way to prove that the Taylor series of sin and cos converge to them. Monotonicity of the integral and the comparison of integrands are basic tools of real analysis, used to bound remainders, errors and limits. Abbott's *Understanding Analysis* develops the integral and Taylor's theorem with remainder carefully.

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